

Numerical inequalities

What “greater than” really means — one definition by subtraction that settles every comparison, however awkward the numbers.

LESSON 38–39

2 × 40 MIN

ALGEBRA 8, §11

STAGE 9 · 4.3

LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- State the definition of $a > b$ by means of the difference.
- Compare two numbers or expressions using the difference method.
- Read and write strict and non-strict inequalities.
- Place numbers correctly on a number line.

TEXTBOOK

National	§11, pp. 68–70
Cambridge	Learner’s Book pp. 96–102
Workbook	Workbook 4.3

01

Explanation

The definition

DEFINITION

$$a > b \text{ means } a - b > 0$$
$$a < b \text{ means } a - b < 0$$

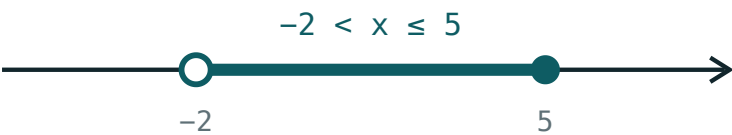
and $a = b$ means $a - b = 0$.

That is the whole of this lesson. Comparing two numbers is not a matter of looking at them — it is a matter of **subtracting and checking the sign**. For ordinary numbers this feels like overkill; for expressions with letters it is the only method that works.

On the number line, $a > b$ means a lies to the **right** of b .

Strict and non-strict

SYMBOL	READ AS	TYPE	ON THE LINE
$a > b$	a is greater than b	strict	open circle
$a < b$	a is less than b	strict	open circle
$a \geq b$	a is greater than or equal to b	non-strict	filled circle
$a \leq b$	a is less than or equal to b	non-strict	filled circle



open circle = strict · filled = included

An open circle excludes the boundary; a filled circle includes it.

Note that $5 \geq 5$ is **true** — “or equal to” only has to hold on one side.

The difference method in action

Compare $\frac{3}{7}$ and $\frac{4}{9}$ without decimals:

$$\frac{3}{7} - \frac{4}{9} = \frac{27 - 28}{63} = -\frac{1}{63} < 0$$

The difference is negative, so $\frac{3}{7} < \frac{4}{9}$. No decimals, no calculator, no rounding to argue about.

With letters it is the only route. To show that $a^2 + 1 > 2a$ for every a , subtract:

$$a^2 + 1 - 2a = (a - 1)^2 \geq 0$$

A square is never negative, so the difference is never negative — the inequality holds, with equality only when $a = 1$.

TWO USEFUL FACTS

$a^2 \geq 0$ for every a , and $a^2 > 0$ unless $a = 0$. Almost every proof in this chapter ends by spotting a square.

02 Worked examples

EXAMPLE 1 Compare $\frac{5}{8}$ and $\frac{7}{11}$

① $\frac{5}{8} - \frac{7}{11}$

Subtract.

② $= \frac{55 - 56}{88} = -\frac{1}{88}$

Common denominator 88.

③ The difference is negative.

④ $\frac{5}{8} < \frac{7}{11}$

ANSWER

$$\frac{5}{8} < \frac{7}{11}$$

EXAMPLE 2 *Prove that $a^2 + 4 \geq 4a$ for every a*

① $a^2 + 4 - 4a$

Take the difference.

② $= a^2 - 4a + 4$

Rearrange.

③ $= (a - 2)^2$

A perfect square.

④ $(a - 2)^2 \geq 0$ always.

So the difference is never negative.

ANSWER

True for every a , with equality at $a = 2$.

EXAMPLE 3 *Which is greater: 2^{10} or 10^3 ?*

① $2^{10} = 1024$

② $10^3 = 1000$

③ $1024 - 1000 = 24 > 0$

The difference is positive.

④ $2^{10} > 10^3$

ANSWER

$$2^{10} > 10^3$$

03 Interactive model

Move the boundary and flip the symbol — the open and filled circles are the whole story.

Solutions on the number line

$x > 2$



inequality $x > 2$

boundary **2 (open)**

$x = 0$ works? **no**

$x = 6$ works? **yes**

04

Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Inequality	Tengsizlik	Неравенство
Greater than	Katta	Больше
Less than	Kichik	Меньше
Strict inequality	Qat’iy tengsizlik	Строгое неравенство
Non-strict inequality	Qat’iy bo‘lmagan tengsizlik	Нестрогое неравенство
Difference	Ayirma	Разность
Positive	Musbat	Положительный
Negative	Manfiy	Отрицательный
Number line	Sonlar o‘qi	Числовая прямая
Compare	Taqqoslash	Сравнить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$a > b$ means:

Which statement is true?

a^2 is:

On the number line, a filled circle means:

the value is excluded

the value is included

the inequality is strict

nothing in particular

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Compare 7 and 4 using the difference

ANSWER $7 - 4 = 3 > 0$, so $7 > 4$

2. Compare -3 and -5

ANSWER $-3 - (-5) = 2 > 0$, so $-3 > -5$

3. Is $6 \geq 6$ true?

ANSWER Yes.

4. Is $6 > 6$ true?

ANSWER No.

5. Write “ x is at least 5” as an inequality

ANSWER $x \geq 5$

6. Write “ x is less than -2 ” as an inequality

ANSWER $x < -2$

7. Compare $\frac{1}{2}$ and $\frac{2}{5}$

ANSWER $\frac{1}{2} > \frac{2}{5}$

Medium · the standard question

7 PROBLEMS

1. Compare $\frac{3}{7}$ and $\frac{4}{9}$

ANSWER $\frac{3}{7} < \frac{4}{9}$

2. Compare $\frac{5}{8}$ and $\frac{7}{11}$

ANSWER $\frac{5}{8} < \frac{7}{11}$

3. Which is greater: 2^{10} or 10^3 ?

ANSWER $2^{10} = 1024 > 1000$

4. Compare $3\sqrt{2}$ and 4

ANSWER $3\sqrt{2} \approx 4.24 > 4$

5. Prove $a^2 + 1 \geq 2a$

ANSWER The difference is $(a - 1)^2 \geq 0$.

6. Prove $a^2 + 4 \geq 4a$

ANSWER The difference is $(a - 2)^2 \geq 0$.

7. Compare $-\frac{2}{3}$ and $-\frac{3}{4}$

ANSWER $-\frac{2}{3} > -\frac{3}{4}$

Hard · stretch and reason

7 PROBLEMS

1. Prove that $a^2 + b^2 \geq 2ab$ for all a, b

ANSWER The difference is $(a - b)^2 \geq 0$.

2. Prove that $a + \frac{1}{a} \geq 2$ for $a > 0$

ANSWER The difference is $\frac{(a - 1)^2}{a} \geq 0$ because $a > 0$.

3. Compare $\frac{a}{b}$ and $\frac{a+1}{b+1}$ for $a > b > 0$

ANSWER The difference is $\frac{a-b}{b(b+1)} > 0$, so $\frac{a}{b} > \frac{a+1}{b+1}$.

4. Prove that $(a + b)^2 \geq 4ab$

ANSWER The difference is $(a - b)^2 \geq 0$.

5. Which is greater: $\sqrt{10}$ or 3.2?

ANSWER $10 < 10.24 = 3.2^2$, so $\sqrt{10} < 3.2$

6. Show that $a^2 + b^2 + 2 \geq 2a + 2b$

ANSWER The difference is $(a - 1)^2 + (b - 1)^2 \geq 0$.

7. For which a is $a^2 > a$?

ANSWER $a^2 - a = a(a - 1) > 0$ when $a < 0$ or $a > 1$.

07 Homework

Homework — 5 problems

Algebra 8, §11, pp. 68–70. Show the difference in every comparison.

- Compare $\frac{4}{9}$ and $\frac{5}{11}$ by the difference method
- Compare $-\frac{3}{5}$ and $-\frac{5}{8}$

3. *Prove that $a^2 + 9 \geq 6a$ for every a*
4. *Which is greater: 3^5 or 5^3 ?*
5. *Write as inequalities: “ x is at most 7” and “ y is greater than -1 ”*